Text Models for Measuring Ideology

POS6933: Computational Social Science

Jake S. Truscott, Ph.D

University of Florida

Spring 2026

Overview

- Housekeeping
- Class 9 Problem Set Review
- `Wordscores`
- `Wordfish`
- `Wordshoal`
- Office Hours (Until 6:00)

Housekeeping

- **Reminders**:
  - Slides due to me and peer reviewers by 11:59pm on Sunday (Schedule *Next*)
  - Final materials (Paper, Appendix, and Final Slides) due 4/24 (Canvas)
  - Rubric & help document located on Canvas (*Files*) – Be sure to follow formatting requirements...
- I will stay until 6:00 today if you want to chat about your papers

Housekeeping – Presentation Order

Day	Student	Reviewer 1	Reviewer 2
4/12	Arora, Manya	Mamun	Revilla, Valeria
4/12	Cui, Lixiao	Meneley, Cory	Sarver, Sarah
4/12	Giron-Khalil, Jailyn	Mohammed, Suren	Sultana, Abida
4/12	Hao, Tianshu	Oneil, Connor	Tierman, Hunter
4/12	Islam, Md Sahidul	Revilla, Valeria	Trenchi, Alejandro
4/12	Mamun	Sarver, Sarah	Wang, Qianxi
4/12	Meneley, Cory	Sultana, Abida	Arora, Manya
4/12	Mohammed, Suren	Tierman, Hunter	Cui, Lixiao
4/19	Oneil, Connor	Trenchi, Alejandro	Giron-Khalil, Jailyn
4/19	Revilla, Valeria	Wang, Qianxi	Hao, Tianshu
4/19	Sarver, Sarah	Arora, Manya	Islam, Md Sahidul
4/19	Sultana, Abida	Cui, Lixiao	Mamun
4/19	Tierman, Hunter	Giron-Khalil, Jailyn	Meneley, Cory
4/19	Trenchi, Alejandro	Hao, Tianshu	Mohammed, Suren
4/19	Wang, Qianxi	Islam, Md Sahidul	Oneil, Connor

Class 9 Problem Set Review

- Notes...

Scaling Ideology

Scaling Ideology (Cont.)

- **How I Was Taught to Think of It**: Rather than using right-hand side of the equation to solve for the left, flip the intuition – Use DV to estimate optimal IV(s).

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  1. **Stable Latent Preferences**: Assumes actors have stable, coherent, underlying ideological positions that guide their behavior.
  2. **Behavior Reflects Ideology**: Treats votes or language as meaningful signals of those latent preferences (not pure noise or strategy).
  3. **Common Dimensional Structure**: Assumes actors operate in a shared ideological space (often low-dimensional) that is comparable across observations.

Scaling Ideology (Goals)

- **Recover Latent Position(s)**: Using observable behaviors as signals (ex: votes on legislation), estimate an underlying ideological dimension that isn't directly observed but we know exists and can be represented given both directionality (e.g., *Left-Right*) and magnitude (e.g., *Very Left-Very Right*).

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- **Place Actors (Groups) On a Spectrum**: Relative location of actors (groups) in a common space denotes cardinality (and proximity) of relationship in that dimension.

Text Models for Ideology (Benefits)

- Binary votes sometimes insufficient to fully understand elite political behaviors
- Text leverages high dimensional structure – more variance and perhaps better explanatory power.
- Text captures nuance in reasoning, framing, and emphasis that votes alone cannot reveal
- Enables dynamic and context-sensitive measurement of ideology across issues, time, and audiences

Models Today

- Going to focus on 3 models: `Wordscores`, `Wordfish`, and `Wordshoal`
- **Important**: View subsequent models as methodological improvements on preceding

Wordscores (Laver, Benoit, and Garry)

- Developed by Laver, Benoit, and Garry (2003) – Probably not(?) first text-based model of ideology, but certainly first to gain significant use in social sciences.
- It's effectively (sort of) Naive Bayes + Value Projection onto a Reference Scale + Rescaling

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  1. Estimate how likely each word w is to appear in each class k – $p(w \mid k)$
  2. For new document D_i, we use those learned word probabilities to calculate π_{ik} (probability that D_i belongs in each possible class – $p(k|midD_i)$
  3. Classification assigned to the document because it has the highest probability

Wordscores (Laver, Benoit, and Garry) – Cont.

- **What `Wordscores` does instead**: Estimate how likely it is that each word belongs to a reference document – $p(w \mid r)$, where A_r represents the ideological score of those reference documents.

Wordscores (Laver, Benoit, and Garry) – Cont.

- **What `Wordscores` does instead**: Estimate how likely it is that each word belongs to a reference document – $p(w \mid r)$, where A_r represents the ideological score of those reference documents.
- $S_w = \sum_r p(r \mid w) A_r$ – the score of some word S_w is the sum of the probabilities it belongs to each reference document times the score associated with those documents.

Wordscores (Laver, Benoit, and Garry) – Cont.

- **What `Wordscores` does instead**: Estimate how likely it is that each word belongs to a reference document – $p(w \mid r)$, where A_r represents the ideological score of those reference documents.
- $S_w = \sum_r p(r \mid w) A_r$ – the score of some word S_w is the sum of the probabilities it belongs to each reference document times the score associated with those documents.
- New documents v are finally rescaled to account for weighted averages of word scores – such that a document's position is the average ideology of the words it uses (from the reference documents) and weighted by the word frequences:

$$S_v = \sum_w p(w \mid v) S_w$$

Wordscores Example

Word	Left Ref.	Right Ref.	TOTAL
welfare	8	1	9
tax	2	6	8
military	1	7	8
TOTAL	11	14	25

- 3 Words (*welfare*, *tax*, *military*)
- 2 reference documents:

$$A_{Left} = (-1)$$
$$A_{Right} = (+1)$$

Wordscores Example (Cont.)

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welfare	8	1	9
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- Let's start with word probabilities:

  $p(\text{left} \mid \text{welfare}) = \frac{8}{9} = 0.889$

  $p(\text{right} \mid \text{welfare}) = \frac{1}{9} = 0.111$

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$p(\text{left} \mid \text{tax}) = \frac{2}{8} = 0.25$

$p(\text{right} \mid \text{tax}) = \frac{2}{8} = 0.75$

$p(\text{left} \mid \text{military}) =$

Wordscores Example (Cont.)

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$p(\text{right} \mid \text{tax}) = \frac{2}{8} = 0.75$

$p(\text{left} \mid \text{military}) = \frac{1}{8} = 0.125$

$p(\text{right} \mid \text{military}) =$

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$$p(\text{right} \mid \text{welfare}) = \frac{1}{9} = 0.111$$
$$p(\text{left} \mid \text{tax}) = \frac{2}{8} = 0.25$$
$$p(\text{right} \mid \text{tax}) = \frac{2}{8} = 0.75$$
$$p(\text{left} \mid \text{military}) = \frac{1}{8} = 0.125$$
$$p(\text{right} \mid \text{military}) = \frac{7}{8} = 0.875$$

Wordscores Example (Cont.)

Word	$p(\text{left} \mid w)$	$p(\text{right} \mid w)$
welfare	0.889	0.111
tax	0.25	0.75
military	0.125	0.875

- Now we can compute the word scores using $S_w = \sum_r p(r \mid w) A_r$:

$$S_{\text{welfare}} = 0.889(-1_{Neg}) + 0.111(1_{Pos}) = -0.778$$

$$S_{\text{tax}}$$

Wordscores Example (Cont.)

Word	$p(\text{left} \mid w)$	$p(\text{right} \mid w)$
welfare	0.889	0.111
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- Now we can compute the word scores using $S_w = \sum_r p(r \mid w) A_r$:

$$S_{\text{welfare}} = 0.889(-1_{Neg}) + 0.111(1_{Pos}) = -0.778$$

$$S_{\text{tax}} = 0.25(-1_{Neg}) + 0.75(1_{Pos}) = 0.50$$

$$S_{\text{military}}$$

Wordscores Example (Cont.)

Word	$p(\text{left} \mid w)$	$p(\text{right} \mid w)$
welfare	0.889	0.111
tax	0.25	0.75
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- Now we can compute the word scores using $S_w = \sum_r p(r \mid w) A_r$:

$$S_{\text{welfare}} = 0.889(-1_{Neg}) + 0.111(1_{Pos}) = -0.778$$

$$S_{\text{tax}} = 0.25(-1_{Neg}) + 0.75(1_{Pos}) = 0.50$$

$$S_{\text{military}} = 0.125(-1_{Neg}) + 0.875(1_{Pos}) = 0.75$$

Wordscores Example (Cont.)

- We can now start computing document scores for new documents $S_v = \sum_w p(w \mid v) S_w$
- Suppose a new document (S_v) contains the following: **Welfare** (3), **Tax** (1), and **Military** (0).
- We know that $p(\text{welfare} \mid v) = \frac{3}{4} = 0.75$, and $p(\text{tax} \mid v) = \frac{1}{4} = 0.25$.
- Solve for S_v

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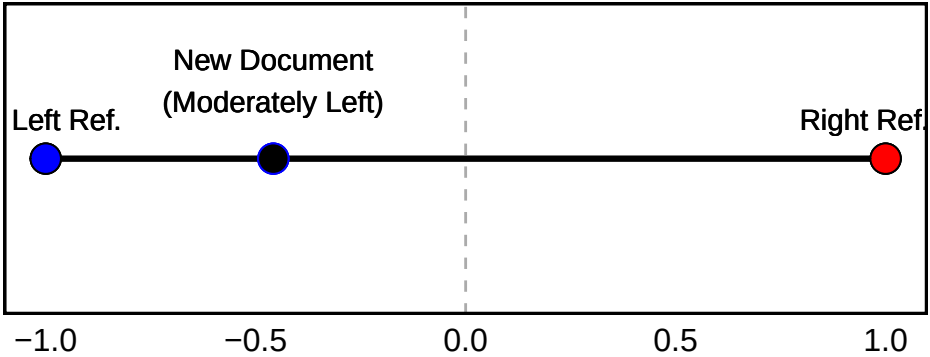
- Suppose a new document (S_v) contains the following: **Welfare** (3), **Tax** (1), and **Military** (0).
- We know that $p(\text{welfare} \mid v) = \frac{3}{4} = 0.75$, and $p(\text{tax} \mid v) = \frac{1}{4} = 0.25$.
- Solve for $S_v = 0.75(-0.778) + 0.25(0.5) = (-0.458)$

Wordscores Example (Cont.)



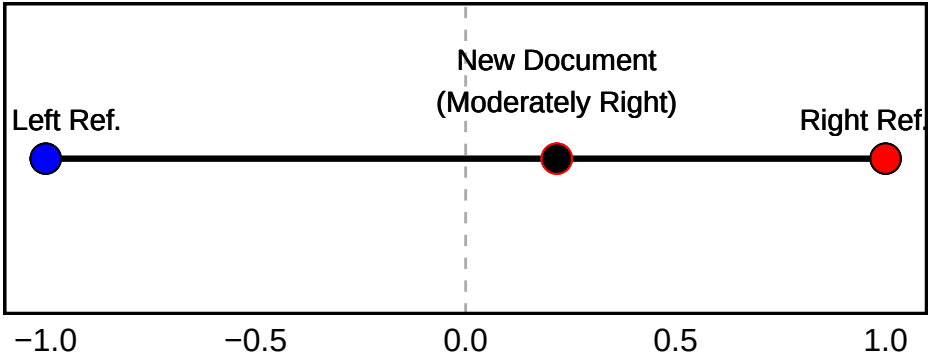
Wordscores Example (Cont.)

- Recall: $S_v = \sum_w p(w \mid v) S_w$
- Suppose another new document (S_v) contains the following: **Welfare** (2), **Tax** (5), and **Military** (3).
- $p(\text{Welfare} \mid v) = 0.20$, $p(\text{Tax} \mid v) = 0.5$, and $p(\text{Military} \mid v) = 0.33$
- Solve for S_v

Wordscores Example (Cont.)

- Recall: $S_v = \sum_w p(w \mid v) S_w$
- Suppose another new document (S_v) contains the following: **Welfare** (2), **Tax** (5), and **Military** (3).
- $p(\text{Welfare} \mid v) = 0.20$, $p(\text{Tax} \mid v) = 0.5$, and $p(\text{Military} \mid v) = 0.33$
- Solve for $S_v = 0.20(-0.778) + 0.5(0.25) + 0.33(0.75) = 0.217$

Wordscores Example (Cont.)



Wordfish (Slapin and Proksch)

- `Wordfish` is a poisson factor model that (like Wordscores) tries to scale the ideological positions of documents.

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- `Wordfish` is a poisson factor model that (like Wordscores) tries to scale the ideological positions of documents.
- It does **not** require reference documents – it is **unsupervised**
- Uses maximum likelihood to infer both a document's latent position θ_i and word discrimination β_j (how strongly a word separates documents)

Wordfish Specification

$$x_{ij} \sim \text{Poisson}(\lambda_{ij})$$

$$\lambda_{ij} = \exp(\alpha_j + \psi_j + \beta_j \theta_i)$$

Where:

- x_{ij} = the count of word j in document i
- α_i = Document-specific verbosity (value to control for length of document – ensures θ_i reflects content differences, not just document size)
- ϕ_j = Word-specific baseline frequency
- β_j = Word-specific discrimination
- θ_i = Latent position of document i (**What we're most interested in**!)